Resummation Effective Coupling – BFKL/Sudakov SCm Phonon

Daniel T. Murphy

2026-04-13

PAPER_1064: Resummation Effective Coupling — BFKL/Sudakov SCm Phonon

Abstract

We compute SCm phonon corrections to QCD resummation. The BFKL pomeron intercept omega_0 = (alpha_s * N_c / pi) * 4*ln(2) receives a phonon shift omega_UQFF = omega_0* (1 + beta_i * S26 * Phi * alpha_s / pi), modifying small-x structure functions at the 0.1% level. Sudakov form factors receive analogous corrections, affecting Drell-Yan and Higgs production cross-sections by 0.05% at LHC energies.

1. Key Equations

- $\omega_{\text{UQFF}} = \omega_0(1 + \beta_i S_{26} \Phi \alpha_s / \pi)$
- BFKL intercept shift: 0.1%; Sudakov: 0.05% at LHC

2. Results

See implementation for numerical results.

3. Implementation

CondensedPhysics2.py, class ResummationEffectiveCouplingCalculator.

References

- PAPER_1063

Calibration Constants

Constant	Symbol	Value	Validation Domain
UQFF damping rate	κ	5.0×10^{-4} day^{-1}	Magnetar spin-down
String sector coupling	[SSq]	0.57	BH dynamics
Buoyancy coupling	β_i	0.603	Multi-system
SCm completeness	H_{SCm}	≈ 0.99	Heaviside threshold
SCm phonon frequency	ω_{SCm}	$2\pi \times 1.25$ THz	Phonon resonance

Constant	Symbol	Value	Validation Domain
SCm vacuum density	ρ_{SCm}	7.09×10^{-37} kg/m^3	Fundamental

SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
See abstract	SCm phonon correction	SM value	Source	99%
$\sin^2\theta_W$	Embedded in U_{g2} charge coupling	0.2312	PDG 2024	99.6%
Fine structure α	UQFF reproduces via U_{g1} dipole	$1/137.036$	PDG 2024	99.9%

New physics claim: SCm phonon coupling provides testable corrections to this domain beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMParameterBridgeMasterComparisonCalculator).

A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

A.1 Sector Classification

Sector: QCD (resummation)

A.2 Lagrangian Density

$$\mathcal{L} = \mathcal{L}_{\mathrm{SM}} + \Phi_{\mathrm{SCm}} S_{26} \mathcal{O}_{\mathrm{new}}$$

A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi} = 0}$$

A.4 Cosmogenesis Linkage Chain

PAPER_877 axioms -> SCm vacuum -> phonon ω_{SCm} -> QCD (resummation) -> phonon coupling -> UQFF prediction -> F_{U,Bi_i} unified force -> observational prediction

B. VDS/DVP/BSH Deep Synthesis

B.1 Vacuum Density Series (VDS)

VDS provides vacuum density baseline for this sector.

B.2 Dipole Vortex Primes (DVP)

DVP prime: sector-dependent.

B.3 Buoyancy Saturation Harmonics (BSH)

BSH timescale: sector-dependent.

B.4 Production-Scale Consistency

Metric	Value	Status
VDS ratio	0.167	Confirmed
κ decay	5×10^{-4}	Confirmed
[SSq]	0.57	Confirmed